

Clinical Risk Assessment Report: Patient 98

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Positive for Metastasis
- Prediction Confidence: High Confidence
- Absolute Predicted Risk: 37.4% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 1.69x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as High-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET), which elevated the patient's absolute risk by +5.6%. Biologically, this feature points to continuous, elongated bands of high-density or highly active tumor tissue.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: 5.57 [Top <1%] (+5.6%)**
Points to continuous, elongated bands of high-density or highly active tumor tissue.

Tumor Pixel Correlation (CT) (GLCM_Correlation.CT) **Value: 1.86 [Top 2%] (+3.7%)**
Measures the linear dependency of gray levels; atypical values corroborate a complex, non-uniform spatial cellular arrangement.

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: 1.49 [Top 5%] (+2.9%)**
Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Large Zone Low Gray-Level Emphasis (CT) (GLZLM_LZLGE.CT) **Value: 2.81 [Top 2%] (+2.7%)**
Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.3 [Avg (25th %ile)] (+2.6%)**
Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: -0.25 [Bottom 5%] (+2.0%)**
Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

Minimum Metabolic Activity (SUVmin) (CONVENTIONAL_SUVbwmin) **Value: 3.7 [Top <1%] (+1.8%)**
Indicates the areas of lowest metabolic activity within the tumor, which may represent necrotic or poorly perfused regions.

3. Protective / Mitigating Factors (Reducing Risk)

Patient Age (Age) **Value: 68.0 [Avg (68th %ile)] (-2.3%)**
Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Machine Learning Feature Attribution (SHAP Decision Path)

